Stellar

Verifying circuits on Stellar

Workshop - Get dangerous under an hour

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THE Aha COMPANY

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Hey 👋 **I am Tupui** nice to meet you.

Come say hi!

@tupui

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The story

001

Welcome ZK

010

The year of ZK

011

DX is key

100

Let’s ZK on Stellar

001 Welcome ZK

Protocol 22 - BLS12-381

```rust
use soroban_sdk::Env;
let env = Env::default();
let bls12_381 = env.crypto().bls12_381();
```

Pedersen commitment

We want to record 2 votes anonymously, we can hide the vote with a random seed

Some ceremony to go from numbers, scalars, to points on the BLS12-381 curve

```
let vote_a = Fr::from_u256(U256::from_u128(&env, 1_u128));
let seed_a = Fr::from_u256(U256::from_u128(&env, 60_u128));
let vote_b = Fr::from_u256(U256::from_u128(&env, 2_u128));
let seed_b = Fr::from_u256(U256::from_u128(&env, 30_u128));
```

```
// generators
let vote_g = Bytes::from_slice(&env, b"VOTE_G");
let vote_dst = Bytes::from_slice(&env, b"VOTE_F");
let seed_g = Bytes::from_slice(&env, b"SEED_G");
let seed_dst = Bytes::from_slice(&env, b"SEED_F");

let vote_g_point = bls12_381.hash_to_g1(&vote_g, &vote_dst);
let seed_g_point = bls12_381.hash_to_g1(&seed_g, &seed_dst);
```

Pedersen commitment

```
// casting votes, on-chain store commitments
// C = g^v * h^r (in additive notation: g*v + h*r)
let vote_point = bls12_381.g1_mul(&vote_g_point, &vote_a);
let seed_point = bls12_381.g1_mul(&seed_g_point, &seed_a);
let commitment_a = bls12_381.g1_add(&vote_point, &seed_point);

let vote_point = bls12_381.g1_mul(&vote_g_point, &vote_b);
let seed_point = bls12_381.g1_mul(&seed_g_point, &seed_b);
let commitment_b = bls12_381.g1_add(&vote_point, &seed_point);

let tally_commitment = bls12_381.g1_add(&commitment_a, &commitment_b);

// collect votes info and prove tally by sending votes and seeds on-chain
let tally_vote_point = bls12_381.g1_mul(&vote_g_point, &(vote_a + vote_b));
let tally_seed_point = bls12_381.g1_mul(&seed_g_point, &(seed_a + seed_b));
let commitment_check = bls12_381.g1_add(&tally_vote_point, &tally_seed_point);

assert_eq!(commitment_check, tally_commitment);
```

010 The year of ZK

Protocol 25

Focus on privacy with indispensable primitives, privacy token…

BN-254

Poseidon and Poseidon2

https://github.com/jayz22/soroban-examples/tree/p25-preview/p25-preview

Ecosystem of verifiers

What is ZK without verifiers!

RISC ZERO

Groph16

NOIR

Barretenberg with Ultrahonk

011 DX is key

3 tools to Quickstart

Stellar CLI

https://developers.stellar.org/docs/tools/cli/install-cli

Open Zeppelin

https://wizard.openzeppelin.com

Scaffold Stellar

https://scaffoldstellar.org

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Let’s ZK

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Thank you

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